CRS Project Brief – Summative Assessment (Individual)

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| **CRS** | Artificial Intelligence |
| **Course Name** | Mathematics forAI-II |
| **Type of Assessment** | Summative Assessment |
| **Marks of Assessment** | 60 Marks |
| **Intended Learning Outcomes Assessed** | • Apply mathematical/statistical logic to interpret data patterns<br>• Process, clean, and structure real-world data using Python<br>• Design interactive, user-focused visualizations<br>• Develop and deploy a working data dashboard using Streamlit Cloud |
| **Assignment title** | **Developing User Centered Dashboards to Solve Real World Problems** |
| **Purpose/Rationale of the Assignment** | This project mimics real-world data analysis tasks where students are expected to uncover insights, create visual summaries, and build intuitive dashboards. Whether it's sports analytics or hospital performance, one must understand client needs, manipulate data, and create tools that guide decisions. Students will use Python (Pandas, NumPy, Matplotlib/Plotly) for analysis and Streamlit to deploy their app |

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| **Scenario or Context** | **Students must select one of the scenarios provided (either Scenario 1: Player Injuries and Team Performance Dashboard or Scenario 2: Hospital Inpatient Discharges Dashboard Based on the chosen scenario, select and complete the corresponding tasks as outlined in the assignment.**<br><br>**Scenario 1: Player Injuries and Team Performance Dashboard**<br><br>FootLens Analytics, a leading sports analytics firm dedicated to performance optimization in football, collaborates with professional leagues and clubs to deliver data-driven insights for improving player performance, minimizing injury risks, and boosting team success. As part of the AI Research & Insights Team at FootLens, you are now tasked with a crucial project. The company has recently acquired a comprehensive dataset detailing player injuries and team performance metrics across several seasons. They aim to uncover how injuries influence a club's overall performance and to derive actionable insights for training schedules, player rotation, and squad planning.<br><br>In your role as a Junior Sports Data Analyst, you are expected to design an interactive dashboard using Streamlit. This dashboard will support technical directors and sports managers in visualizing and interpreting the connection between player injuries, match outcomes, and team standings, ultimately guiding data-backed decision-making on the field and beyond.<br><br>**OR** |

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| | **Scenario 2: Hospital Inpatient Discharges Dashboard**<br><br>MediScope Health Analytics, a leading healthtech company specializing in AI-driven healthcare solutions, has partnered with hospitals to enhance patient care and streamline hospital operations through data insights.<br><br>As a member of MediScope's Healthcare AI and Data Insights Team, you have been given access to a comprehensive dataset on hospital inpatient discharges across multiple departments. Your objective is to examine trends in hospital admissions and discharges, analyze length of stay patterns, identify departments with high readmission rates, and extract actionable insights that can contribute to operational efficiency and improved patient outcomes.<br><br>You are a Data analyst tasked with creating an interactive dashboard using Streamlit. This dashboard will support hospital administrators, clinicians, and operations managers in visualizing critical metrics and making well-informed, data-backed decisions. |

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| **Task 1** | **Task 1: Player Injuries and Team Performance Dashboard**<br>**Step 1: Understand the Problem & Research**<br>**Dataset to be used from <Drive link>**<br><br>Begin by exploring the dataset thoroughly. Understand the key columns like player details, match results, injury dates, and team performance ratings. Research what sports analysts look for, such as how injuries affect a team's overall performance and the role of star players. Use your research to frame five meaningful business questions that will drive the rest of your analysis.<br>● Review the dataset structure: player name, team, match result, injury start/end date, performance rating<br>● Understand what metrics matter to analysts (e.g., player impact, match outcomes, post-recovery performance)<br>● Conduct background research on sports analytics (performance index, injury severity)<br>● Formulate 5 key questions for your reference (students can formulate and analyse their own at least 5):<br><br>○ Which injuries led to the biggest team performance drop?<br>○ What was the team's win/loss record during player absence?<br>○ How did individual players perform after recovery?<br>○ Are there specific months or clubs with frequent injury clusters?<br>○ Which clubs suffer most due to injuries?<br><br>**Step 2: Data Preprocessing and Cleaning**<br>After understanding the problem, the next step is to clean and preprocess the data. This includes handling missing values, converting dates, and engineering new columns that quantify player performance drop and injury impact. Cleaned data ensures your analysis and visualizations are accurate and relevant.<br>● Load the dataset using `pandas`<br>● Handle missing values (NaNs) in columns like ratings, goals, and injury dates<br>● Convert injury start and end dates to `datetime` format<br>● Create new columns:<br><br>○ Player's average rating before and after injury |

- Team's performance drop index during player absence

- Rename unclear column names for better readability
- Group data by `Player Name` to calculate summary stats across injury phases (before, during, after)

**Step 3: Exploratory Data Analysis**

With clean data in hand, explore the dataset to extract valuable insights. Perform statistical summaries and visual grouping to reveal trends like most frequently injured players or the effect of age on recovery. These insights will inform your visualizations and help validate your research questions.

- Use groupby() and agg() to find:
  - Players with the most injuries
  - Clubs with frequent injury cases
  - Players who showed major improvement or decline after injury
- Identify top 5 injuries with highest impact on team performance (goal difference or ratings)
- Use `.pivot_table()` to compare metrics like pre- and post-injury performance
- Use statistical summaries to calculate recovery trends across players

**Step 4: Build Interactive Visualizations (Minimum 5 Visuals)**

Convert your insights into **5 compelling, interactive visualizations**. These will showcase the most impactful data stories like which players made the strongest comebacks or when injury spikes occurred. Ensure clarity, relevance, and interactivity in all visuals.

- Bar Chart: Top 10 injuries with highest team performance drop
- Line Chart: Player performance timeline (before and after injury)
- Heatmap: Injury frequency across months and clubs
- Scatter Plot: Player age vs. performance drop index
- Leaderboard Table: Comeback players ranked by rating improvement

**Step 5: GitHub & Streamlit Deployment**

Create a GitHub repo, write a clear README explaining what your app does, then deploy to Streamlit Cloud so anyone can try it. Good documentation and sharing show pride in your work.

- **What to do ?:**

➔ Create requirements.txt listing needed Python packages (like Streamlit, pandas).
➔ Create a GitHub repository (name it clearly).
➔ Upload app.py (your main code) and requirements.txt.
➔ Go to Streamlit Cloud:

  ➢ Click "Deploy an app".
  ➢ Connect to your GitHub.
  ➢ Select your repo and deploy for free.

➔ Write a clear README including:

  ➢ What your dashboard analyses.
  ➢ Key features.
  ➢ Live link and screenshots.

**Note:** Upload the final Live streamlit web app link to the GitHub repository including a detailed README file. Repository name : eg: IADAI102(Student_id)-studentname.

**Task 2: Hospital Inpatient Discharges Dashboard**

**Step 1: Understand the Problem & Research**

Dataset to be used from **<Drive link>**

Start by understanding the dataset fields such as patient demographics, diagnosis codes, hospital charges, and severity level. Research how hospital administrators use such data to optimize planning and reduce costs. Frame 5 research questions that will guide your entire analysis.

- Study the dataset: age, gender, diagnosis codes, charges, length of stay, hospital region
- Research typical healthcare analysis topics (cost by severity, regional differences, age-based charges)
- Develop 5 key questions for your reference ( students can formulate and analyse their own at least 5):
    - Which diagnosis codes are linked to long stays?
    - Which age groups incur the highest charges?
    - How does patient severity affect total cost?
    - Are there counties with longer average stays?
    - How do different payment types affect the final billing?

**Step 2: Data Preprocessing and Cleaning**

Clean the hospital data to make it usable for analysis. Convert string and currency formats, remove outliers, and transform categorical variables into readable formats. These steps are essential for clear analysis and visualization.

- Load data using `pandas`
- Clean or remove invalid fields like negative stay duration
- Convert monetary values (charges) from strings to `float`
- Standardize categorical fields:
    - Age group, gender, race, severity
- Map diagnosis codes to readable medical categories
- Drop irrelevant or null-heavy columns like birth weight (if mostly empty)

**Step 3: Exploratory Data Analysis**

Perform in-depth analysis to uncover insights about patient discharge trends. Use aggregation functions and visual summaries to find out which diagnoses, demographics, or facilities have the highest cost or stay durations.

- Group by diagnosis and severity to analyze average stay
- Study total hospital charges by age group and gender
- Identify top high-utilization facilities
- Use `pivot_table()` to uncover patterns across regions and payment types
- Visualize outliers or trends in severity and cost distribution

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| | **Step 4: Build Interactive Visualizations (Minimum 5 Visuals)**<br><br>Design 5 key visualizations that answer your research questions. These visuals should offer clarity for non-technical audiences and support decision-making. Make your charts interactive and insightful.<br><br>● Bar Chart: Avg. hospital stay per diagnosis code<br>● Box Plot: Total charges segmented by severity<br>● Heatmap: Average stay duration by facility and county<br>● Pie Chart: Distribution of patients by payment type<br>● Histogram: Overall distribution of length of stay<br><br>Hint: Use libraries like `Plotly`, `Altair`, or `Seaborn` for interactive visuals<br><br>**Stage 5: Deploy & document your project**<br><br>Finish by making your project accessible and professional. Publish your code on GitHub, write a README that explains your design and logic, and deploy on Streamlit Cloud so anyone can use it online.<br><br>● **What to do ?:**<br><br>➔ Create requirements.txt listing needed Python packages (like Streamlit, pandas).<br>➔ Create a GitHub repository (name it clearly).<br>➔ Upload app.py (your main code) and requirements.txt.<br>➔ Go to Streamlit Cloud:<br><br>➢ Click “Deploy an app”.<br>➢ Connect to your GitHub.<br>➢ Select your repo and deploy for free.<br><br>➔ Write a clear README including:<br><br>➢ What your dashboard analyses.<br>➢ Key features.<br>➢ Live link and screenshots.<br><br>**Note:** Upload the final Live streamlit web app link to the GitHub repository including a detailed README file. Repository name : eg: IADAI202(Student_id)-studentname. |
| **Checklist of evidence required** | ● **Create a submission document (.pdf) and upload it when submitting the Project Assignment.**<br><br>Make sure your document includes:<br>**1. Link to GitHub**<br>➔ Go to your GitHub repository page.<br>➔ Provide access to ai.assignments@wacpinternational.org<br>➔ Copy the full URL from the address bar (e.g., https://github.com/YourUsername/YourRepository)<br>➔ Paste the GitHub link directly into the document where you want it to appear, or create a hyperlink by selecting the text, right-clicking, and choosing the "Insert Link" or "Hyperlink" option. Paste the URL in the link field.<br><br>**2. Student’s Full Name**<br>**3. Candidate Registration Number** |

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| | **4. CRS Name : Artificial Intelligence**<br>**5. Course Name**<br>**6. School Name**<br><br>• **GitHub Repository checklist:**<br><br>(Ensure the repository is functional and accessible.)<br><br>• **Create below Folders and upload deliverables as shown:**<br><br>• **Interactive Links:**<br><br>➢ Live web app URLs included directly in the README or in GitHub repository.<br><br>• **Provide a detailed README file with below details:**<br><br>➢ Project overview<br><br>➢ Integration details<br><br>➢ Deployment instructions<br><br>➢ Live web app/dashboard streamlit cloud link |
| **Assessment Criteria: 60 Marks (Rubrics) - "Appendix - A"** | |
| **Sources of information to support you with this Assignment** | • **Useful Reference for Player Injuries and Team Performance Dashboard**<br><br>➔ https://medium.com/@atharvtembhurnikar/football-analysis-using-streamlit-and-plotly-3d0eaed2ac5b<br>➔ https://towardsdatascience.com/visualise-serie-a-football-league-data-with-pandas-and-plotly-3d26bcf26f2d/<br>➔ https://link.springer.com/article/10.1007/s44196-023-00217-6<br>➔ https://towardsdatascience.com/tag/sports-analytics/<br>➔ https://www.pencilandpaper.io/articles/ux-pattern-analysis-data-dashboards<br>➔ https://www.askpython.com/python-modules/pandas/groupby-to-group-dataframe-by-month<br>➔ https://docs.streamlit.io/get-started/fundamentals/main-concepts<br>➔ https://docs.python.org/3/library/turtle.html<br>➔ https://docs.streamlit.io/develop/api-reference/media/st.image<br><br>• **Useful References for Hospital Inpatient Discharges Dashboard**<br><br>➔ https://blog.streamlit.io/improving-healthcare-management-with-streamlit/<br>➔ https://plotly.com/python/plotly-express/<br>➔ https://python.plainenglish.io/create-60-mind-blowing-graphs-charts-maps-animations-with-plotly-a9695d06bfe1<br>➔<br>➔ https://docs.streamlit.io/develop/api-reference/layout<br>➔ https://medium.com/@richardhightower/streamlit-part-2-layouts-components-and-graphs-8232ef352ed1<br>➔ https://docs.streamlit.io/develop/api-reference/status<br>➔ https://discuss.streamlit.io/t/change-the-progress-bar-color/8189<br>➔ https://www.datacamp.com/tutorial/turtle-graphics |

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| | **To create streamlit app via github repository:**<br>https://www.youtube.com/watch?v=JL9xOs-G1hI<br><br>**To create github repository and readme file:**<br>https://www.youtube.com/watch?v=rCt9DatF63I |

# Appendix A - RUBRICS:

### Assessment Criteria: (Rubrics)

| Criteria | Rubrics to be referred | Maximum Marks | Marks scored |
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| Understanding the Problem | Table 1 | 10 | |
| Data Preprocessing and Cleaning | Table 1 | 10 | |
| Exploratory Data Analysis and Insight Extraction | Table 1 | 15 | |
| Dashboard Visualization and Design | Table 1 | 15 | |
| GitHub Repository and Streamlit Deployment | Table 1 | 10 | |
| **Total** | | **60** | **/60** |

### Table 1

| Criteria | Distinguished (5) | Proficient (4) | Apprentice (3) | Novice (2) | Marks |
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| **Understanding the Problem** (10 Marks) | Demonstrates deep understanding of the domain and dataset; framed 5+ insightful and domain-relevant research questions | Shows clear understanding of the dataset; 4–5 relevant research questions provided with basic research. | Shows partial understanding; 3–4 questions framed, some lack relevance or clarity. | Limited or incorrect understanding; fewer than 3 questions or unclear relevance to the problem. | |

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| | with evidence of external research. | | | | |
| **Data Preprocessing and Cleaning** (10 marks) | Data fully cleaned and transformed; advanced preprocessing (e.g., feature engineering, correct formatting); code is modular, clean, and well-commented. | Data mostly cleaned with appropriate preprocessing steps; code is functional and understandable. | Basic data cleaning done; limited feature engineering or code lacks clarity. | Data cleaning incomplete or incorrect; major issues in code or processing logic. | |
| **Exploratory Data Analysis and Insight Extraction** (15 marks) | Strong insight generation using effective statistical summaries and visual EDA; clear connection to research questions. | Relevant insights presented; EDA supports most research questions. | Basic analysis completed; insights are somewhat connected to the problem. | EDA is superficial or disconnected from research goals. | |
| **Dashboard Visualization and Design** (15 marks) | 5+ interactive, visually compelling and accurate charts; dashboard is intuitive and addresses user needs. | Minimum 5 charts used with interactivity and relevance; meets purpose. | Some interactivity or clarity are missing; charts are present but may lack purpose or design quality. | Poorly designed or minimal visualizations; lack of clarity or interactivity. | |
| **GitHub Repository and Streamlit Deployment** (10 Marks) | Repository is complete with structured folders, clear README, working live link, and screenshots; deployment successful. | Repo mostly complete with working app and README; minor issues in structure or details. | GitHub and deployment are partially complete; the app works with some issues or README lacks clarity. | GitHub repo missing key files or app not deployed; README absent or vague. | |
| **TOTAL** | | | | | **/60** |